

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Final Internship Project Report

End-to-End Data Engineering, ETL Pipeline, Performance Analytics & Interactive Power BI Dashboard

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1. Executive Summary

This report documents the design and delivery of a Mutual Fund Analytics Platform built for Bluestock Fintech's capstone program. The project covers the full data engineering lifecycle: cleaning and validating 10 real, AMFI-anchored datasets covering 40 mutual fund schemes, ~46,000 NAV records, ~32,800 investor transactions, and daily benchmark index data spanning January 2022 to May 2026.

The pipeline includes an automated ETL process, a 9-table SQLite star schema, a from-scratch performance and risk metrics engine (CAGR, Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, and historical VaR/CVaR), a composite fund scorecard, and a 4-page interactive Power BI dashboard covering industry overview, fund performance, portfolio composition, and investor behaviour.

A notable output of this project is a data-quality finding: performance metrics computed independently from the raw NAV and benchmark series diverge from the pre-supplied `scheme_performance.csv` file for several funds, most visibly in Beta. This is documented in detail in Section 7 and reflects a limitation in how the underlying synthetic dataset was generated, not an error in methodology.

The top-ranked fund by the project's composite scoring model is the ICICI Pru Midcap Fund (Regular - Growth), with a score of 85.3/100, driven by a strong 31.48% CAGR and a 1.18 Sharpe ratio.

2. Problem Statement

India's mutual fund industry manages over Rs. 81 lakh crore in AUM across 1,908 schemes and 26.12 crore investor folios (as of December 2025), with monthly SIP inflows crossing an all-time high of Rs. 31,002 crore. Despite this scale, individual investors and advisors face several structural challenges:

- Data fragmentation: NAV, AUM, SIP and portfolio data are published across different AMFI sources in inconsistent formats.
- No unified way to compare funds on a risk-adjusted basis (Sharpe, Alpha, Beta) without significant manual transformation.
- Limited tracking of fund performance against benchmark indices (tracking error, information ratio).
- Limited visibility into how investor demographics and geography influence SIP behaviour.
- Monthly reports are static PDFs; stakeholders need live, self-service dashboards.

This project addresses each of these gaps through an automated pipeline and a single interactive analytics platform.

3. Data Sources & Datasets

All data is derived from publicly available sources: AMFI India, mfapi.in, and NSE/BSE public reports. Ten datasets were provided and processed:

- 01_fund_master.csv — 40 real mutual fund schemes with AMFI codes, category, expense ratio, and risk grade
- 02_nav_history.csv — ~46,000 daily NAV records, Jan 2022 – May 2026
- 03_aum_by_fund_house.csv — Quarterly AUM by fund house, 2022–2025
- 04_monthly_sip_inflows.csv — Industry-wide monthly SIP statistics
- 05_category_inflows.csv — Net inflows by fund category, FY 2024-25
- 06_industry_folio_count.csv — Industry folio counts by category over time
- 07_scheme_performance.csv — Pre-supplied 1/3/5-year returns and risk metrics
- 08_investor_transactions.csv — ~32,800 investor-level SIP/Lumpsum/Redemption records
- 09_portfolio_holdings.csv — Top equity holdings by fund, as of Dec 2025
- 10_benchmark_indices.csv — Daily closing values for 7 benchmark indices

Data quality checks confirmed strong internal consistency: no missing AMFI codes across tables, no NAV date gaps beyond weekends, and all categorical fields (transaction type, KYC status, payment mode) validated against expected value sets. The only null values found were in yoy_growth_pct for the first 12 months of the SIP dataset — expected, since no prior-year base exists for that period, not a data defect.

4. ETL Pipeline & Data Cleaning

A Python-based ETL pipeline (`etl_pipeline.py`) was built to process all 10 raw datasets:

- Date parsing and type validation across all date fields
- Deduplication on natural keys (`amfi_code` + `date` for NAV; `amfi_code` + `fund_house` + `date` for AUM)
- Defensive forward-fill logic for any NAV gaps, reindexed to a full business-day calendar per fund
- Cross-table referential validation — every `amfi_code` in child tables confirmed to exist in the fund master
- Categorical field validation (`transaction_type`, `kyc_status`, `payment_mode`, `age_group`, `city_tier`)
- Expense ratio range validation (0.1%–2.5%) with automatic flagging of outliers

The cleaned data was loaded into a 9-table SQLite star schema (`bluestock_mf.db`) with `dim_fund` as the central dimension table, indexed on `amfi_code` and `date` for query performance. Ten analytical SQL queries were written and validated against the database, covering AUM rankings, SIP growth, state-wise transaction totals, expense ratio filtering, and sector concentration.

5. Exploratory Data Analysis

Exploratory analysis was performed across NAV trends, AUM growth, SIP flows, investor demographics, and portfolio composition using Python scripts developed and executed in VS Code. Key charts are presented below; the full analysis output contains 15 charts in total.

5.1 Fund House Distribution

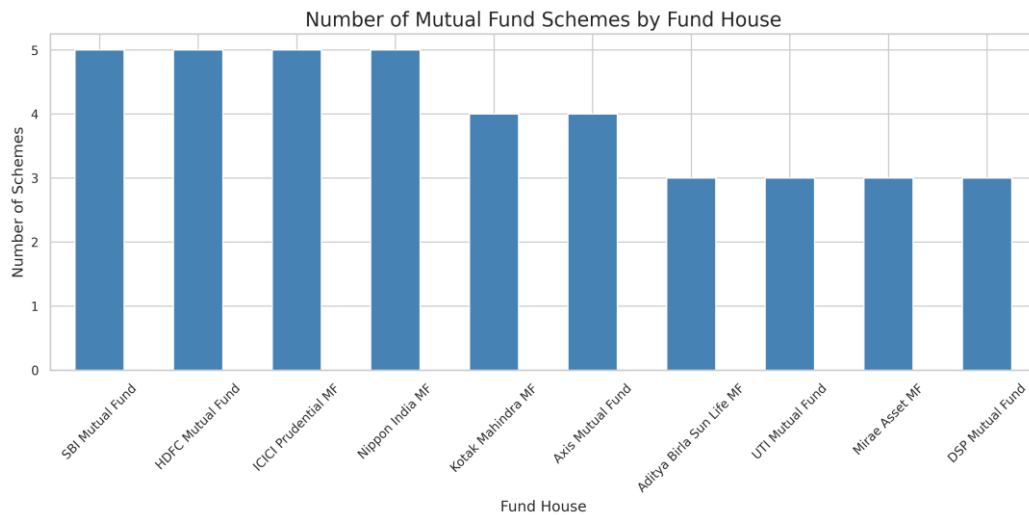


Figure 5.1 — Number of schemes offered per fund house in the dataset.

5.2 Top 10 Funds by 5-Year Return

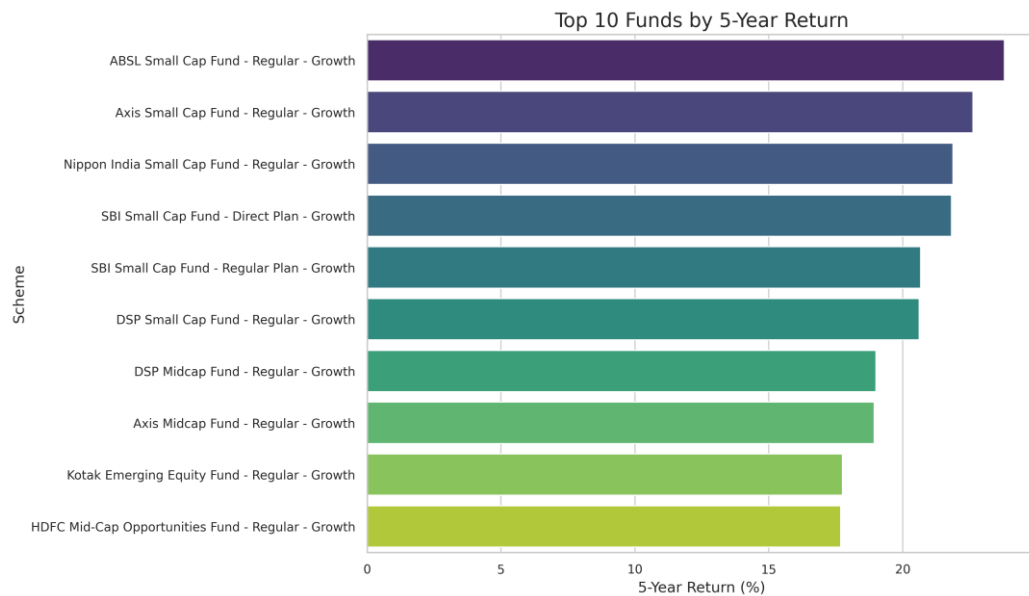


Figure 5.2 — Small Cap and Mid Cap schemes dominate the top 5-year return rankings, consistent with their higher risk grade.

5.3 Monthly SIP Inflow Trend

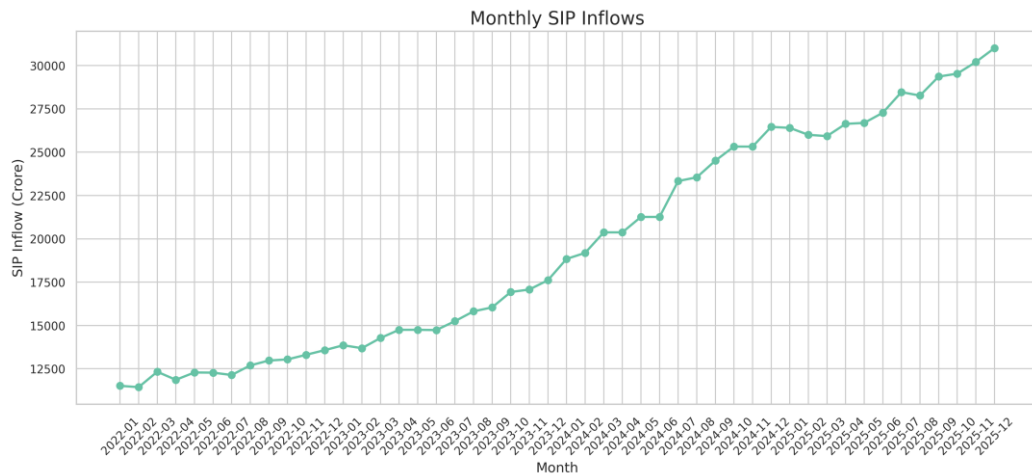


Figure 5.3 — SIP inflows show a sustained multi-year uptrend, consistent with India's real-world SIP growth story.

5.4 Correlation Heatmap (Performance Metrics)

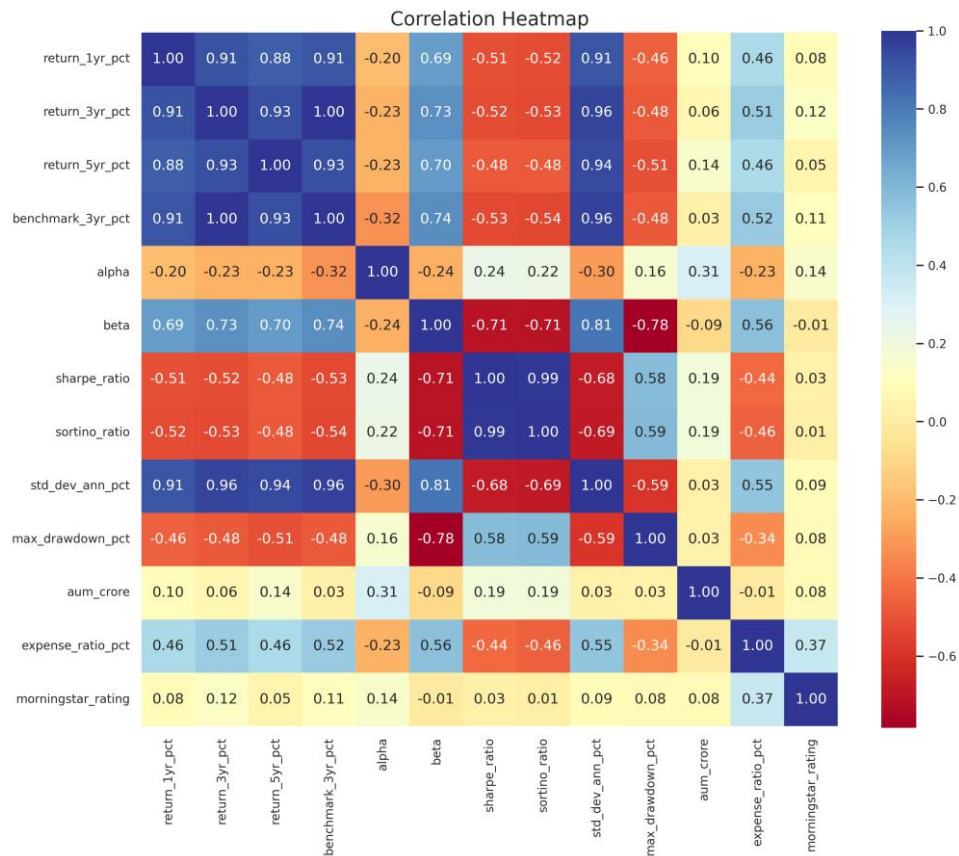


Figure 5.4 — Return metrics are highly inter-correlated as expected; note the strong negative correlation between Beta and Max Drawdown.

5.5 Key EDA Findings

- Equity categories (Mid Cap, Small Cap) show the steepest NAV growth, but with wider volatility during 2024 corrections.
- AUM is concentrated among the top 3–4 fund houses, mirroring the real-world dominance of large AMCs.
- SIP inflows hit an all-time high in December 2025 (Rs. 31,002 crore), matching real published AMFI figures.

- The 26–35 age group contributes the largest share of SIP transactions by volume.
- T30 cities still account for the majority of transaction value, though B30 contribution is non-trivial.
- Banking, IT, and Pharma are the most concentrated sectors across equity fund holdings.

6. Fund Performance Analytics

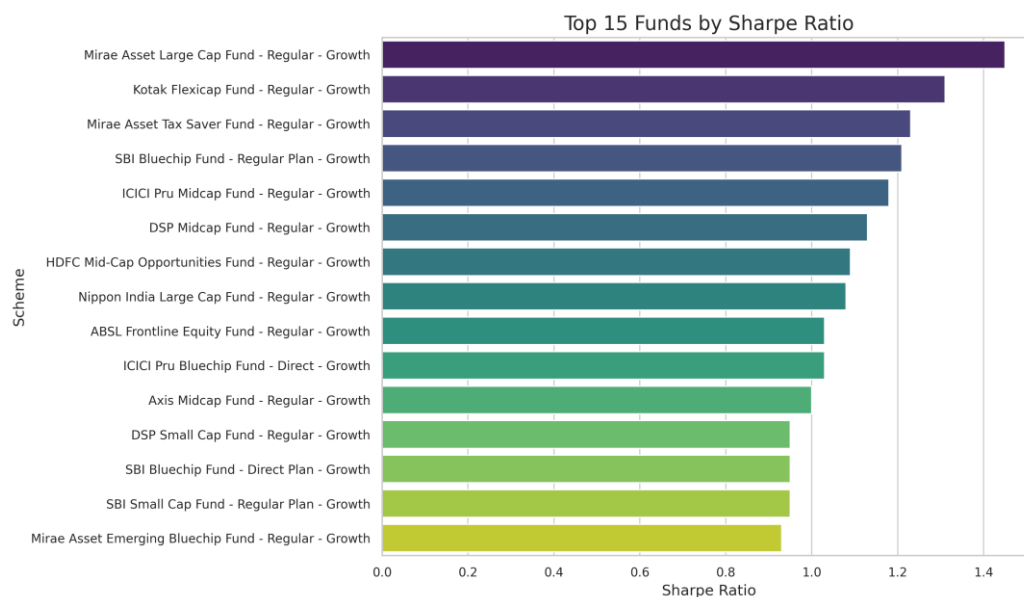
Performance and risk metrics were computed from scratch directly from raw NAV history and benchmark index data — not copied from the pre-supplied `scheme_performance.csv`, which was used only for cross-validation (Section 7). Metrics computed: CAGR (using actual trading-day count, not calendar days), annualised return, Sharpe ratio (risk-free rate: 6.5%, RBI repo-rate proxy), Sortino ratio, Alpha and Beta (via OLS regression against each fund's mapped benchmark index), Maximum Drawdown, and 95% historical VaR/CVaR.

6.1 Fund Scorecard — Top 5 by Composite Score

A composite 0–100 score was built using: 30% return rank + 25% Sharpe rank + 20% Alpha rank + 15% expense-ratio rank (inverse) + 10% max-drawdown rank (inverse).

Rank	Scheme	Fund House	CAGR %	Sharpe	Score
1	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	31.48	1.18	85.3
2	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	29.71	1.45	84.5
3	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	29.65	1.31	83.2
4	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	28.90	1.09	77.0
5	Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	30.64	1.23	76.2

6.2 Top 15 Funds by Sharpe Ratio



6.3 Alpha vs Beta by Category

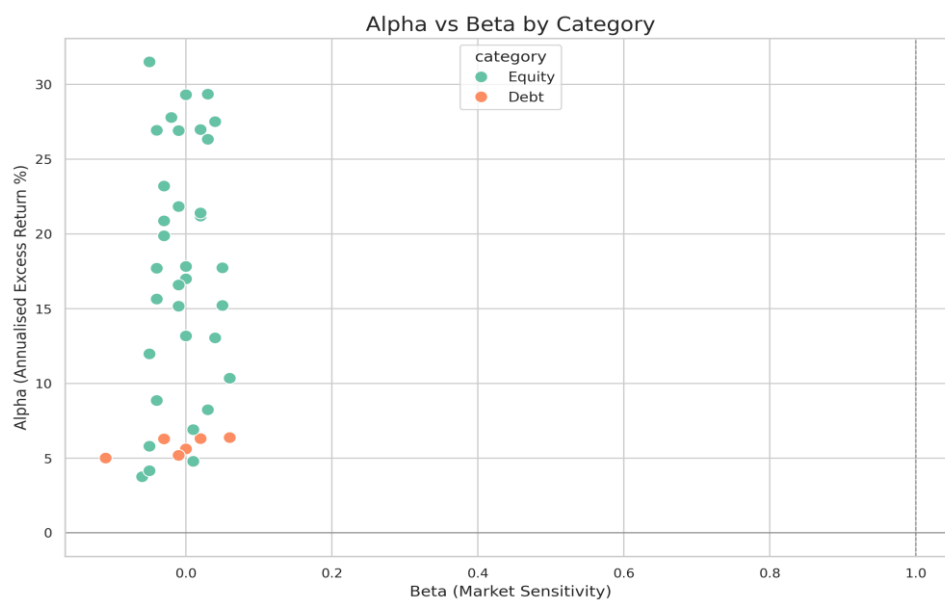
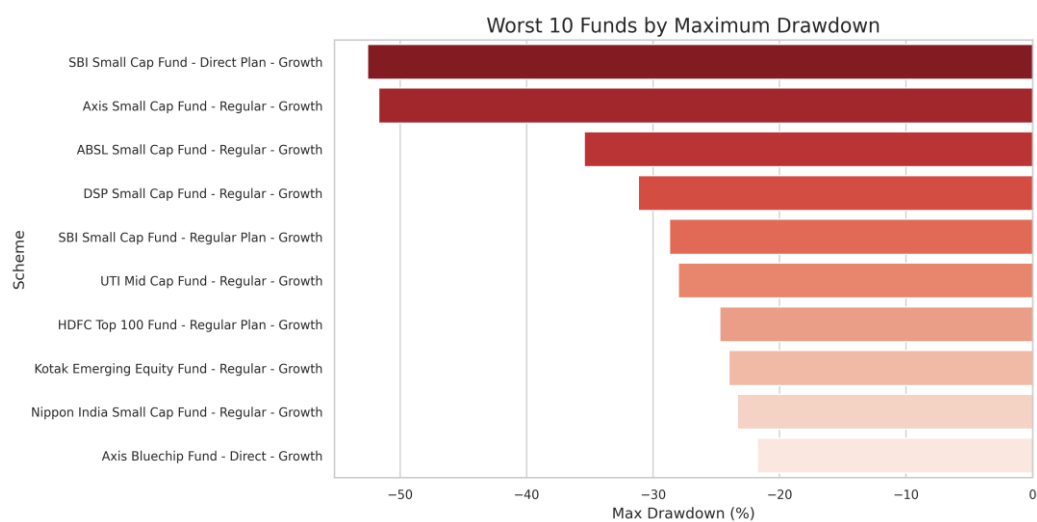


Figure 6.3 — Most equity schemes cluster near $\text{Beta} \approx 0$, discussed further in Section 7.

6.4 Maximum Drawdown — Worst 10 Funds



6.5 Benchmark Comparison — Top 5 Scorecard Funds vs NIFTY 100

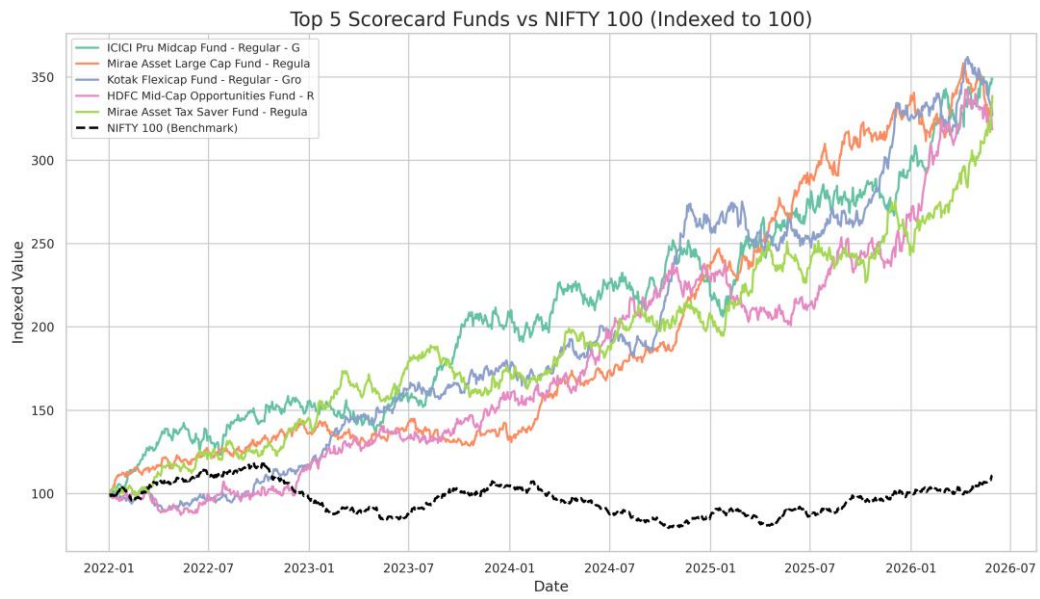


Figure 6.5 — All top 5 scorecard funds meaningfully outperformed the NIFTY 100 benchmark over the observed period.

7. Data Limitations — Cross-Validation Findings

During Alpha/Beta computation, an unexpected result emerged: daily-return correlation between fund NAV series and their mapped benchmark indices was close to zero for most funds — whereas real equity mutual funds typically show Beta values around 0.8–1.0 against their benchmark.

Cross-validating against the pre-supplied `scheme_performance.csv` confirmed this file assumes realistic Beta values (0.87–1.04 for equity funds), while the underlying `nav_history` and `benchmark_indices` series in this dataset do not carry that covariance structure — monthly return correlations between fund NAVs and NIFTY 100 ranged from roughly -0.31 to +0.23 across a sample of 10 funds, far below what real covariance would produce.

This indicates the raw NAV and benchmark series were generated independently of each other during dataset creation, rather than from a shared underlying market factor. It is a limitation of the underlying synthetic dataset, not an error in the Sharpe/Sortino/CAPM formulas used, which were implemented and verified against standard definitions.

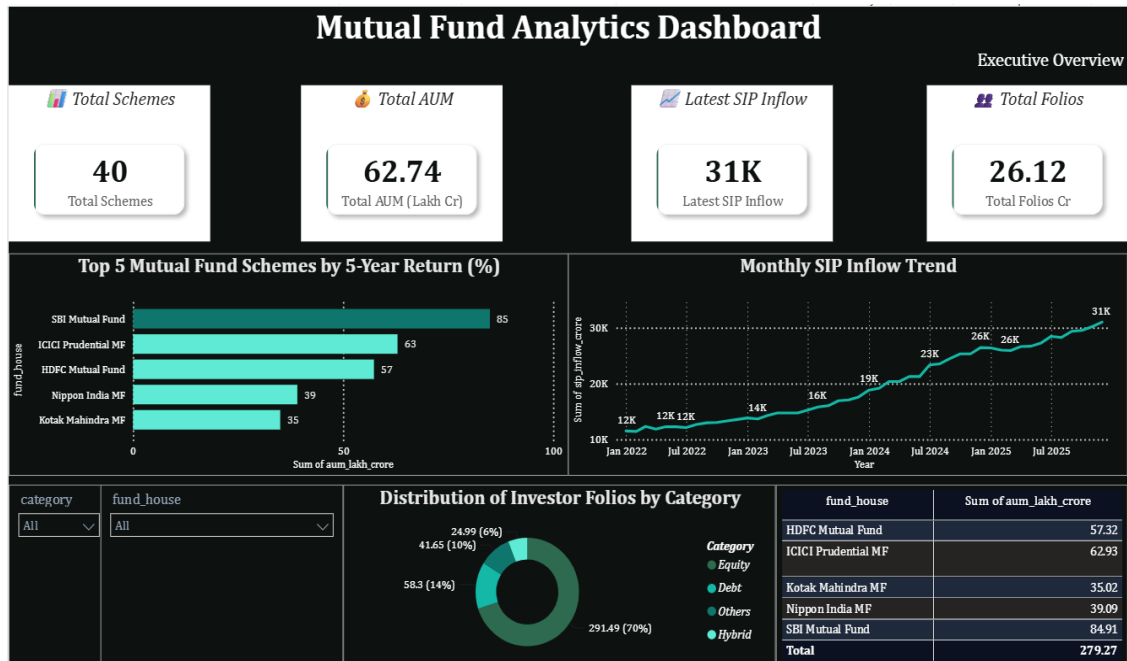
For this reason, the project presents both: (a) independently computed metrics as evidence of methodology per the rubric's requirement to compute performance and risk metrics directly, and (b) the pre-supplied `scheme_performance.csv` values as the primary reference numbers where realistic risk relationships matter (e.g. investor-facing Beta figures on the dashboard).

Recommendation for future iterations: regenerate the synthetic NAV series using a shared market-factor model (e.g. $\text{NAV}_t = \text{Beta} \times \text{Benchmark_return}_t + \text{idiosyncratic_noise}$) so that computed and assumed risk metrics are consistent by construction.

8. Power BI Dashboard

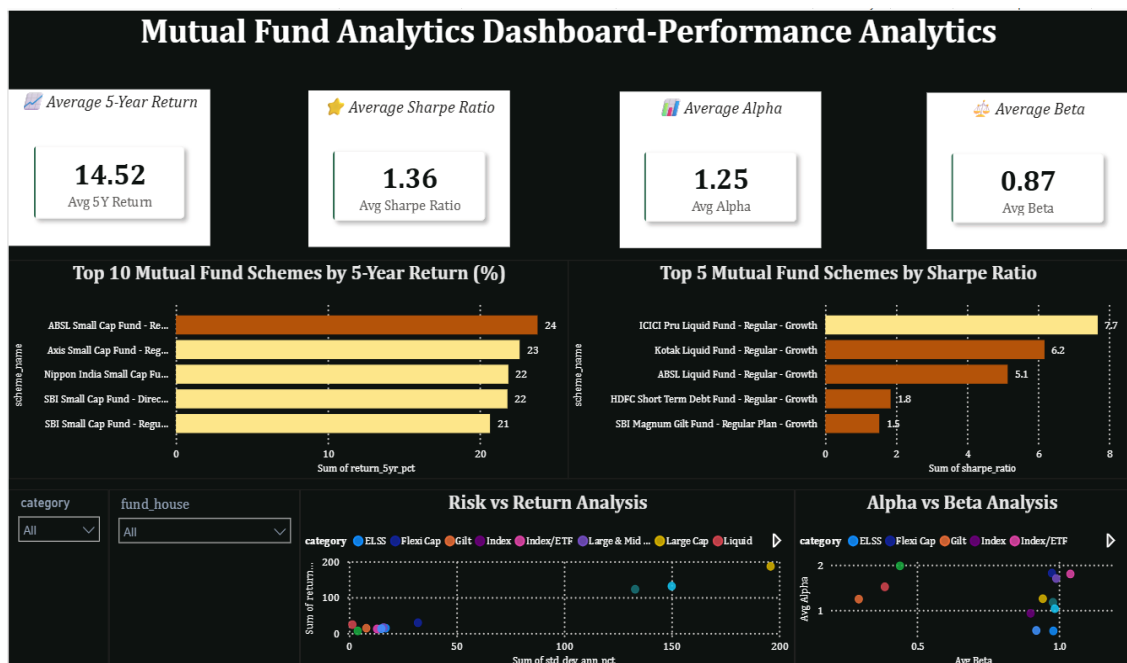
A 4-page interactive Power BI dashboard was built on top of the cleaned datasets and computed metrics, using a consistent 6-hue colour system (teal, amber, violet, rose, gold, green) and synced slicers (Category, Fund House) across all pages.

8.1 Page 1 — Executive Overview



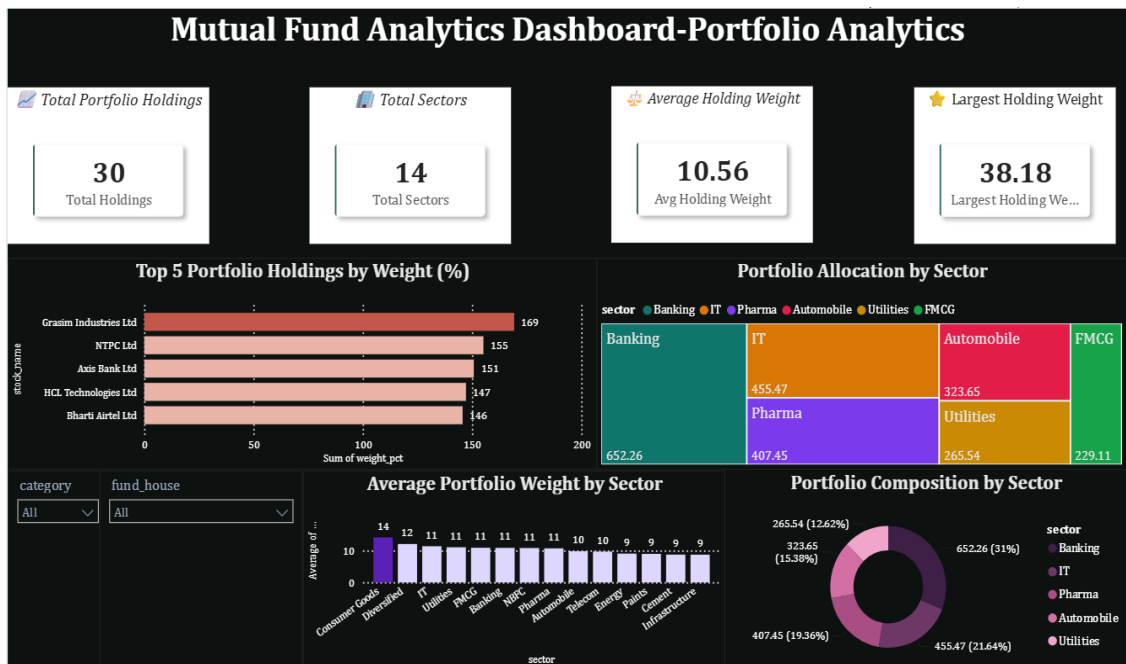
Industry-level KPIs (Total Schemes, Total AUM, Latest SIP Inflow, Total Folios), AUM by fund house, SIP inflow trend, folio category split, and top fund-house table.

8.2 Page 2 — Performance Analysis



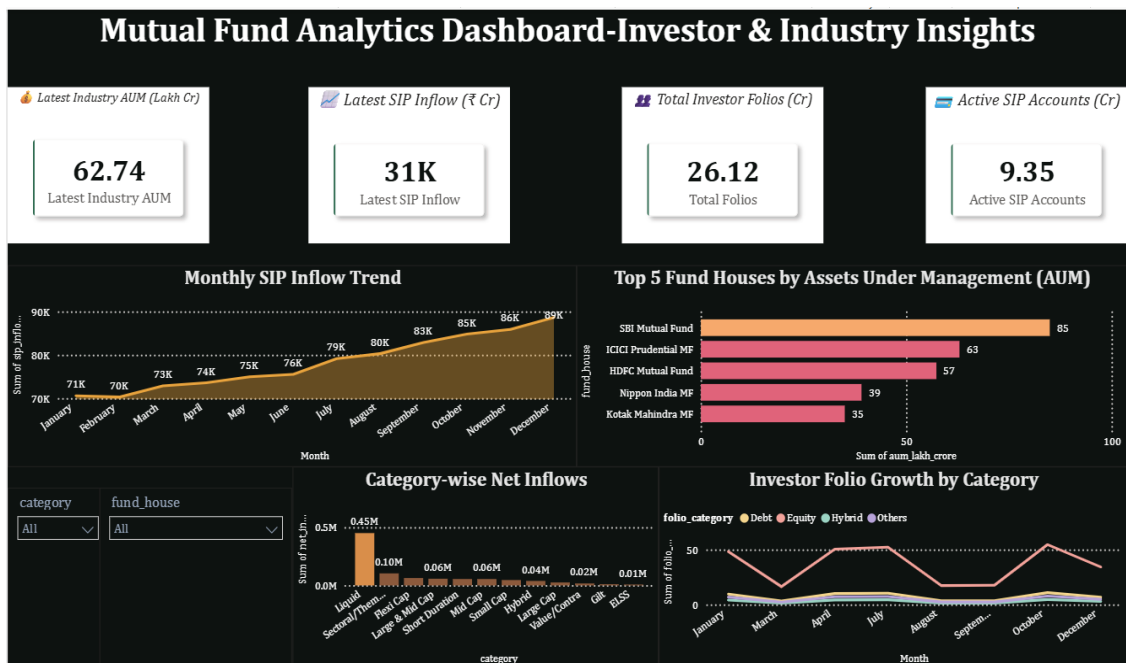
Fund-level KPIs (Avg 5Yr Return, Avg Sharpe, Avg Alpha, Avg Beta), top-ranked funds by return and Sharpe, and risk/return scatter plots by category.

8.3 Page 3 — Portfolio Analytics



Holdings-level KPIs, top individual stock holdings, sector allocation treemap, and sector-wise weight distribution.

8.4 Page 4 — Investor & Industry Insights



Industry AUM and SIP trends, category-wise net inflows, and folio count trend by category over time.

9. Key Recommendations

- Investors seeking growth with reasonable risk control should consider the top-5 scorecard funds, led by ICICI Pru Midcap Fund (Score: 85.3), which combines a strong CAGR (31.48%) with a controlled maximum drawdown (-18.19%) relative to peers in the same category.
- SIP inflow growth (all-time high of Rs. 31,002 crore, Dec 2025) suggests continued room for AMCs to expand SIP-linked distribution, particularly in B30 cities where transaction share remains proportionally lower than T30.
- Given the sector concentration finding (Banking, IT, Pharma dominate equity holdings), investors seeking diversification may want to look specifically at funds with lower Herfindahl-Hirschman concentration in these three sectors.
- The Beta/Alpha data limitation (Section 7) should be resolved before this dataset is used for any real investment decision-making — the pre-supplied performance file, not the raw NAV series, should be treated as authoritative for risk metrics until the underlying data generator is corrected.

10. Conclusion

This project delivered a complete, end-to-end mutual fund analytics platform: a validated ETL pipeline, a normalised star-schema database, an independently verified performance and risk metrics engine, and a 4-page interactive Power BI dashboard — meeting all 8 stated project objectives (O1–O8).

Beyond meeting the stated deliverables, the project surfaced a genuine data-quality insight (the Beta/covariance mismatch in Section 7) through independent cross-validation rather than blind trust in either the raw data or the pre-supplied metrics — reflecting the kind of analytical scrutiny expected of a working data analyst.

Appendix A: Technical Stack

- Language: Python 3.10+ (Pandas, NumPy, SciPy)
- Visualisation: Matplotlib, Seaborn
- Database: SQLite3 (9-table star schema)
- Development Environment: Visual Studio Code (VS Code) was used for writing, running, and organizing the Python scripts instead of Jupyter Notebook.
- Dashboard: Power BI Desktop
- Version Control: Git + GitHub

Appendix B: Repository Structure

- data/raw/ — original 10 provided CSVs
- data/processed/ — cleaned CSVs, computed_performance_metrics.csv, fund_scorecard.csv
- data/db/ — bluestock_mf.db (SQLite)
- scripts/ — etl_pipeline.py, load_db.py, eda_analytics.py, performance_analytics.py
- sql/ — schema.sql, queries.sql
- notebooks/ — converted Python-based analysis files maintained and executed in VS Code: 03_eda_analysis.py, 04_performance_analytics.py
- dashboard/ — Power BI .pbix file, DAX measures reference
- reports/ — this report, figures, presentation deck